NORTHWIND TRADE

Interactive Sales Analysis using Power BI

Project Report on Sales, Product, Customer and Employee Performance

Submitted by

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1. Introduction

Northwind Trade is a well-known sample trading company that imports and exports speciality food products to customers around the world. Its data covers customers, employees, products, categories, orders, order details and shippers, which makes it an ideal case for practising business intelligence and data analytics.

This report presents an interactive sales dashboard built in Microsoft Power BI Desktop for Northwind Trade. The dashboard converts raw transactional data into clear visuals and key performance indicators (KPIs) so that management can quickly understand total sales, order volume, freight cost, customer spread and employee performance. For a student of FinTech with AI, this kind of analysis is a foundation for data-driven decision making, financial reporting and, later, predictive modelling.

2. Objectives of the Project

- **Measure overall performance:** total sales, total orders, quantity sold and freight cost.
- **Identify top categories and products:** find which product groups generate the most revenue.
- **Understand the customer base:** analyse the number of customers by country.
- **Evaluate employee contribution:** compare total orders handled by each sales employee.
- **Build an interactive dashboard:** allow slicing by country, category and order date, with tooltip pages for quick KPI previews.

3. Dataset and Data Model

The dataset used is the Northwind database, loaded into Power BI as seven related tables. Relationships between the tables (for example orders to order_details, and products to categories) allow measures such as Total Sales and Total Orders to be calculated across any dimension.

Table	Description
categories	Product category names (Beverages, Dairy Products, Confections, etc.)
customers	Customer details including company and country
employees	Sales employees who handle the orders
orders	Order header information such as order date, employee, customer and freight
order_details	Line items of each order: product, unit price and quantity
products	Product names linked to their categories
shippers	Shipping companies used to deliver orders

3.1 Key Measures Created

Measure	Purpose
Total Sales	Sum of unit price × quantity (after discount) across all order lines
Total Orders	Count of distinct orders

Measure	Purpose
Total Quantity Sold	Sum of quantity across all order lines
Total Freight	Sum of freight charges on orders
Average Freight	Average freight cost per order
Number of Customers	Distinct count of customers

4. Tools and Methodology

- **Data loading and cleaning:** tables were imported and checked in Power Query (Transform data).
- **Data modelling:** relationships were set between fact and dimension tables.
- **Measures:** DAX measures were written for the KPIs listed above.
- **Visualisation:** card visuals, line chart, column chart, bar charts, a table and slicers were used.
- **Design:** a dark blue themed background was applied across all four report pages for a consistent look.

The final report contains four pages: Sales Overview, Product & Customer Analysis, Interactive Sales Dashboard and Tool Tips.

5. Page 1: Sales Overview

The first page gives a quick summary of the business using six KPI cards and three supporting charts.



Figure 1: Sales Overview page in Power BI

5.1 Key Performance Indicators

KPI	Value	Remark
Average Freight	79.05	Per order
Number of Customers	91	Distinct customers
Total Freight	63.96K	All orders
Total Orders	809	Orders placed
Total Quantity Sold	51K (51,317)	Units
Total Sales	1.27M (1,265,793.04)	Revenue

5.2 Total Orders by Month

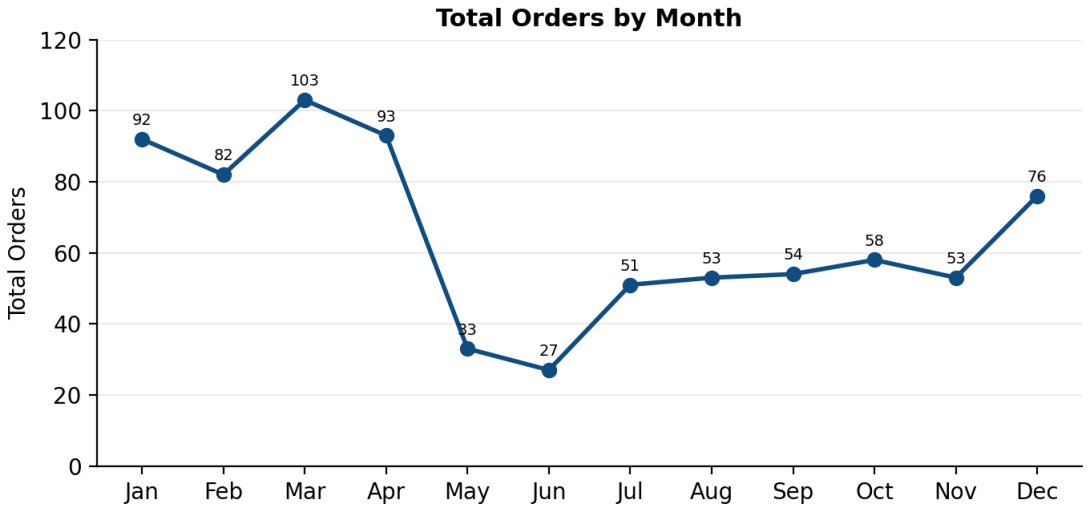


Figure 2: Monthly order trend (values read approximately from the dashboard line chart)

Order volume is highest in the first quarter, peaking around March at roughly 100 orders. There is a sharp fall in May and June, which are the weakest months, followed by a partial recovery in the second half of the year, with a rise again in December.

5.3 Total Sales by Category

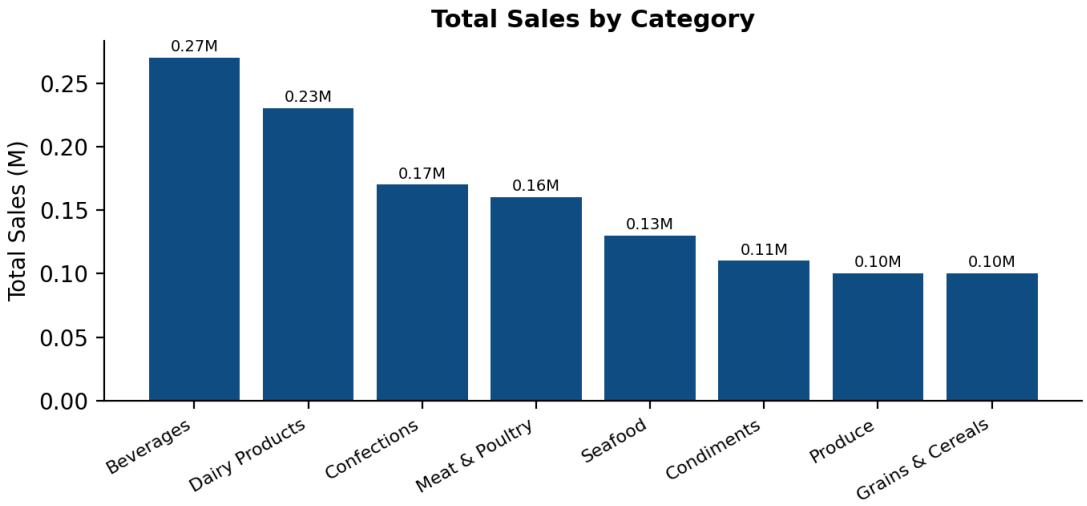


Figure 3: Total sales by product category (approximate values)

Beverages is the top-selling category at about 0.27M, followed by Dairy Products (about 0.23M) and Confections (about 0.17M). Produce and Grains & Cereals contribute the least, at about 0.10M each.

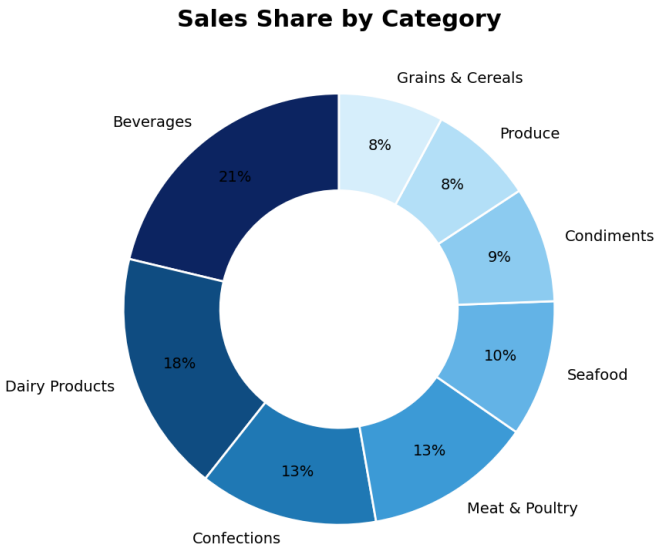


Figure 4: Share of total sales by category

5.4 Total Orders by Employee

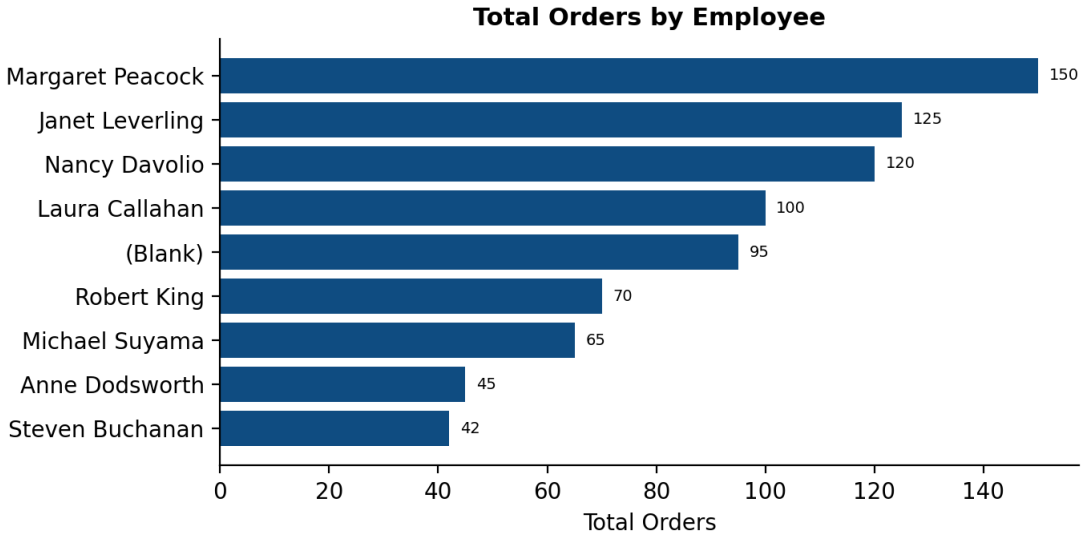


Figure 5: Orders handled by each employee (approximate values)

Margaret Peacock handled the most orders (about 150), followed by Janet Leverling and Nancy Davolio. A (Blank) employee entry of roughly 95 orders also appears, which indicates orders with no employee assigned in the data.

6. Page 2: Product & Customer Analysis

The second page looks at who the customers are and which products drive sales.

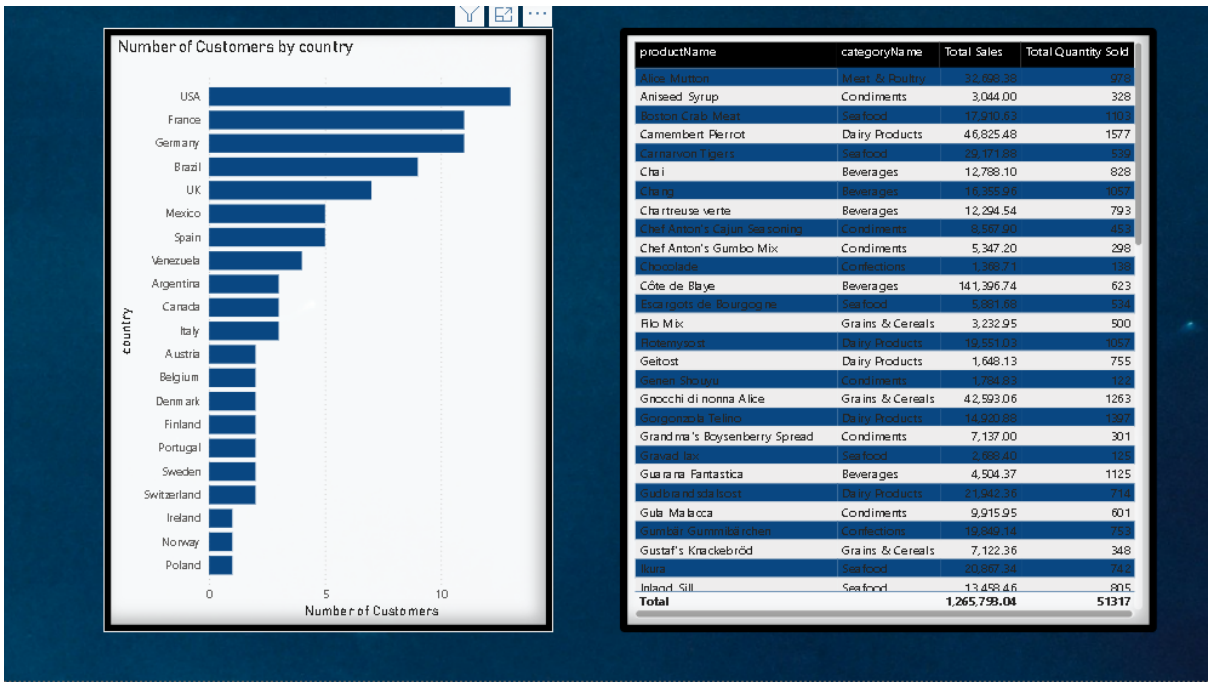


Figure 6: Product & Customer Analysis page in Power BI

6.1 Number of Customers by Country

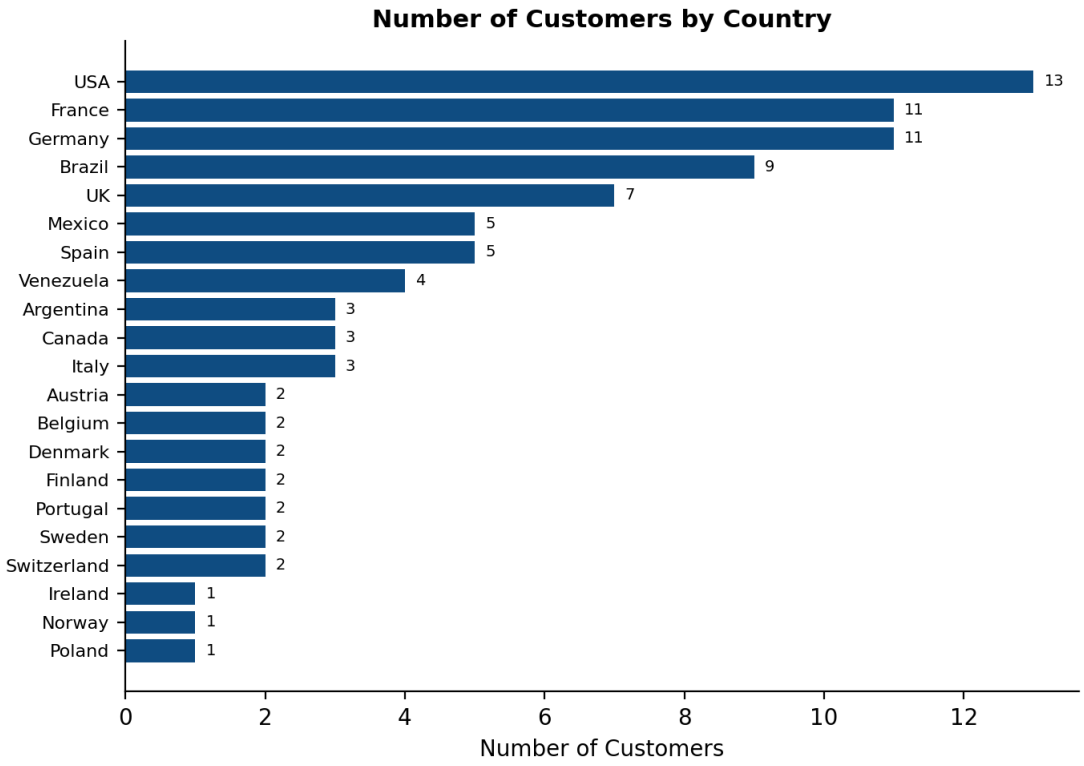


Figure 7: Customers by country (91 customers in total)

The USA has the most customers (13), followed by France and Germany (11 each) and Brazil (9). These four countries together account for 44 of the 91 customers. Ireland, Norway and Poland have only one customer each.

6.2 Product Sales Table

The table below lists total sales and total quantity sold by product and category, as shown on the dashboard (first 29 products alphabetically).

Product Name	Category	Total Sales	Quantity Sold
Alice Mutton	Meat & Poultry	32,698.38	978
Aniseed Syrup	Condiments	3,044.00	328
Boston Crab Meat	Seafood	17,910.63	1,103
Camembert Pierrot	Dairy Products	46,825.48	1,577
Carnarvon Tigers	Seafood	29,171.88	539
Chai	Beverages	12,788.10	828
Chang	Beverages	16,355.96	1,057
Chartreuse verte	Beverages	12,294.54	793
Chef Anton's Cajun Seasoning	Condiments	8,567.90	453
Chef Anton's Gumbo Mix	Condiments	5,347.20	298
Chocolade	Confections	1,368.71	138
Côte de Blaye	Beverages	141,396.74	623
Escargots de Bourgogne	Seafood	5,881.68	534
Filo Mix	Grains & Cereals	3,232.95	500
Flotemysost	Dairy Products	19,551.03	1,057
Geitost	Dairy Products	1,648.13	755
Genen Shouyu	Condiments	1,784.83	122
Gnocchi di nonna Alice	Grains & Cereals	42,593.06	1,263
Gorgonzola Telino	Dairy Products	14,920.88	1,397
Grandma's Boysenberry Spread	Condiments	7,137.00	301
Gravad lax	Seafood	2,688.40	125
Guaraná Fantástica	Beverages	4,504.37	1,125
Gudbrandsdalsost	Dairy Products	21,942.36	714
Gula Malacca	Condiments	9,915.95	601
Gumbär Gummibärchen	Confections	19,849.14	753
Gustaf's Knäckebröd	Grains & Cereals	7,122.36	348
Ikura	Seafood	20,867.34	742
Inlagd Sill	Seafood	13,458.46	805
Ipoh Coffee	Beverages	23,526.70	580
Grand Total (all products)		**1,265,793.04**	**51,317**

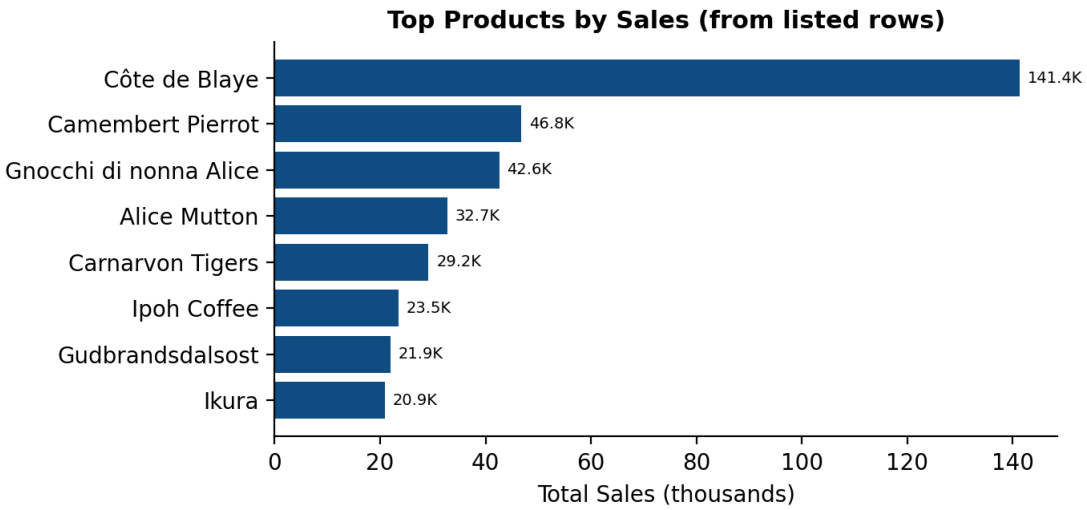


Figure 8: Highest-selling products among the rows listed above

Côte de Blaye is the standout product with sales of about 141K, roughly 11% of total company sales from a single item, even though only 623 units were sold. This shows it is a high-value product. Camembert Pierrot and Gnocchi di nonna Alice follow, with sales of about 46.8K and 42.6K.

7. Page 3: Interactive Sales Dashboard

The third page, titled Northwind Interactive Sales Dashboard, brings the KPI cards and the main charts together on one screen and adds slicers so that the viewer can filter the data.

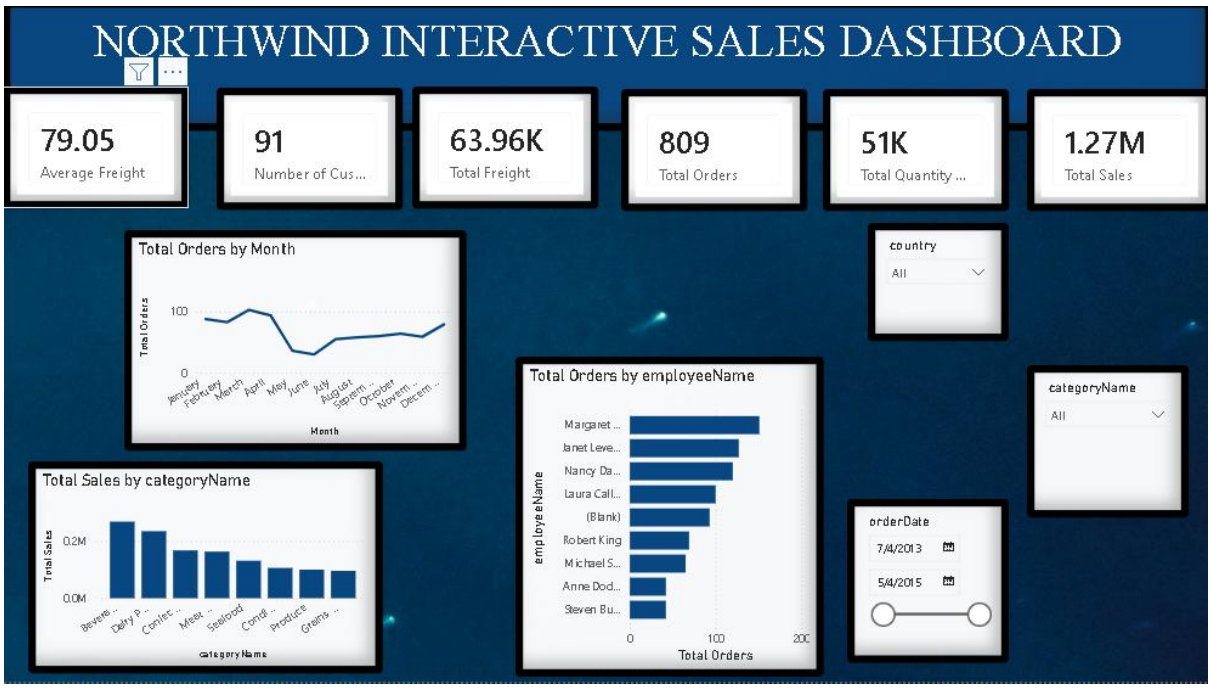


Figure 9: Northwind Interactive Sales Dashboard

7.1 Slicers Used

Slicer	How it helps
country	Filters every visual to show data for a chosen country
categoryName	Shows sales and orders for one product category at a time
orderDate	Date range slider to analyse any time period

Because Keep all filters is switched on, selecting a value in one visual or slicer automatically updates the KPI cards, monthly orders, category sales and employee orders. This makes the dashboard useful for quick what-if style questions, such as how many orders a particular employee handled in a given country.

8. Page 4: Tool Tips

The fourth page is a tooltip page. It holds three compact cards (Total Orders 809, Total Sales 1.27M and Quantity Sold 51K) that appear as a small pop-up when the user hovers over a visual on the other pages. This gives extra context without crowding the main report.



Figure 10: Tooltip page design

9. Key Findings

Metric	Result
Total sales	1.27 million (1,265,793.04)
Total orders	809
Average sales per order	about 1,565 (1,265,793 ÷ 809)
Average quantity per order	about 63 units
Average orders per customer	about 8.9 (809 ÷ 91)
Freight as share of sales	about 5% (63.96K ÷ 1.27M)
Top category	Beverages (about 21% of sales)
Top three categories	Beverages, Dairy Products and Confections (about 53% of sales)
Top product listed	Côte de Blaye (about 141K sales)
Top country by customers	USA (13), then France and Germany (11 each)
Top employee by orders	Margaret Peacock (about 150 orders)

- **Seasonality:** orders fall sharply in May and June and peak in the first quarter.
- **Concentration:** a few categories and one product contribute a large share of revenue.
- **Customer spread:** about 38% of customers are in the USA, France and Germany.
- **Data quality:** a (Blank) employee value shows that some orders are missing an employee assignment.

10. Recommendations

- Plan promotions and marketing campaigns for May and June to lift the low-order months.
- Protect the supply and pricing of Beverages and Dairy Products, as they drive over 40% of sales.
- Review stock and pricing of high-value items such as Côte de Blaye to reduce dependence on a single product.
- Increase customer acquisition in countries with few customers, such as Ireland, Norway and Poland.
- Fix the missing employee records so that performance can be measured fairly.
- Monitor freight cost per order to keep it at or below its current level of about 5% of sales.

11. Scope for FinTech and AI

The dashboard can be extended using AI and financial analytics techniques. Examples include forecasting monthly sales with time-series models, customer segmentation and churn prediction using machine learning, detecting unusual orders or freight charges using anomaly detection, and using Copilot in Power BI to ask questions of the data in plain language. These extensions connect directly with the skills studied in B.Com FinTech with AI.

12. Conclusion

The Northwind Trade interactive sales dashboard successfully turns transaction data into useful business insight. With total sales of about 1.27 million from 809 orders and 91 customers, the report shows where revenue comes from, how it changes through the year, who the key customers and employees are, and where the business can improve. The use of KPI cards, slicers and tooltips makes the report easy to read and interactive, and it demonstrates practical Power BI, DAX and data storytelling skills.

Note: *Chart values for monthly orders, category sales and employee orders were read from the dashboard visuals and are approximate. KPI values and the product table are taken directly from the dashboard.*